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# Digital Image Fundamentals

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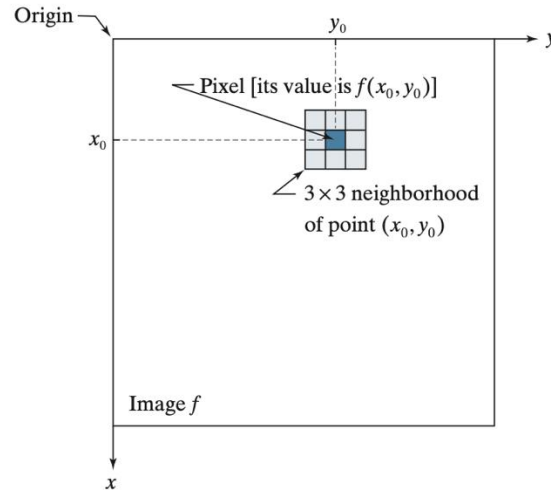
The slide features a light blue background. At the top, there are two horizontal bars: a thick teal one and a thin blue one. At the bottom, there are also two horizontal bars: a thin blue one and a thick teal one. Two short, thick, olive-green horizontal dashes are positioned on the left and right sides of the slide, below the main title.

# The Basics Of Intensity Transformations

The spatial domain processes we discuss in this chapter are based on the expression

$$g(x, y) = T[f(x, y)]$$

where  $f(x, y)$  is an input image,  $g(x, y)$  is the output image, and  $T$  is an operator on  $f$  defined over a neighborhood of point  $(x, y)$ .

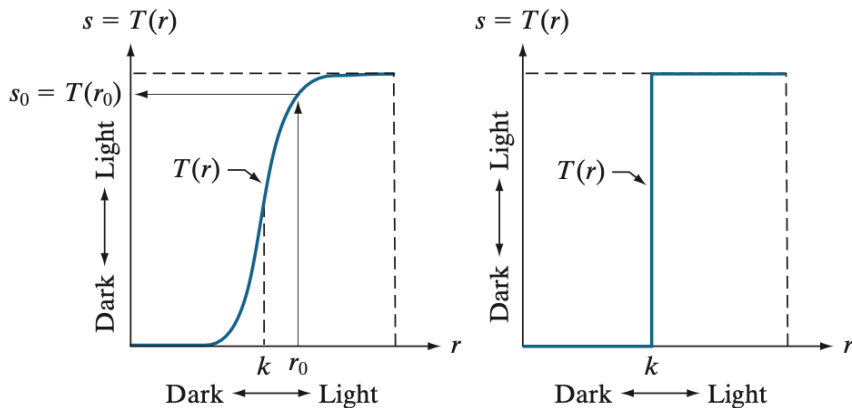


# The Basics Of Intensity Transformations

If  $g$  depends only on the value of  $f$  at a single point  $(x, y)$ ,  $T$  becomes an *intensity* (also called a *gray-level*, or *mapping*) *transformation function* of the form

$$s = T(r)$$

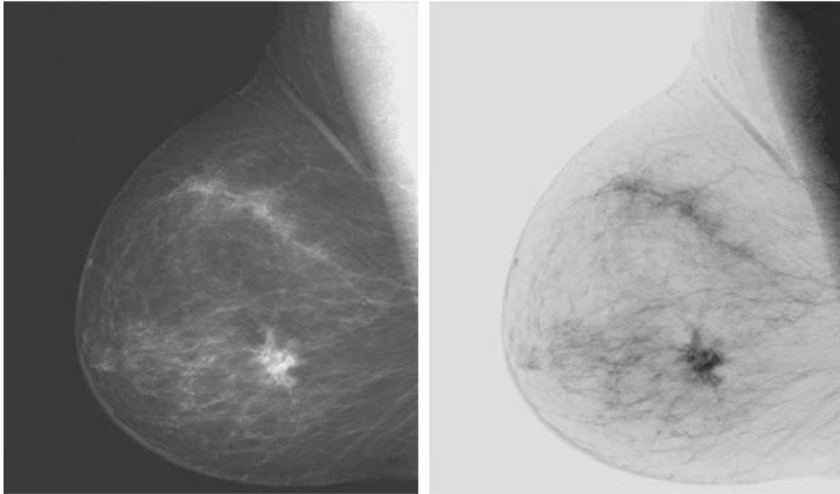
where, for simplicity in notation, we use  $s$  and  $r$  to denote, respectively, the intensity of  $g$  and  $f$  at any point  $(x, y)$



# Image Negatives

The negative of an image with intensity levels in the range  $[0, L - 1]$  is obtained by using the negative transformation function

$$s = L - 1 - r$$



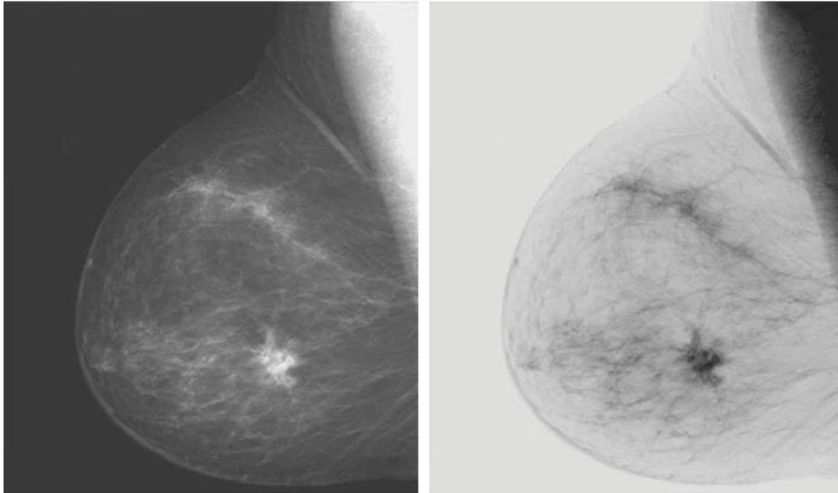
(a) A digital mammogram.

(b) Negative image, converted from image (a)

# Image Negatives

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(a) A digital mammogram.

(b) Negative image, converted from image (a)

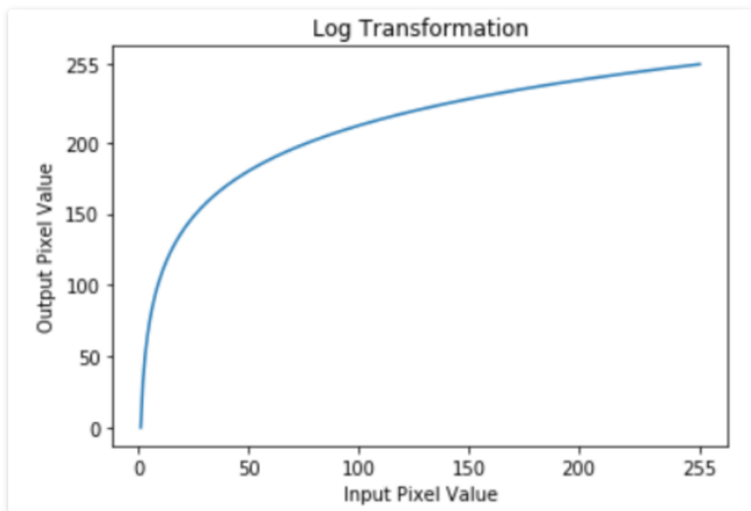
In grayscale images, when the background is black, the foreground gray levels are not clearly visible. So, converting background to white, the gray levels now become more visible.

# Log Transformations

The general form of the log transformation is

$$s = c \log(1 + r)$$

where  $c$  is a constant and it is assumed that  $r \geq 0$

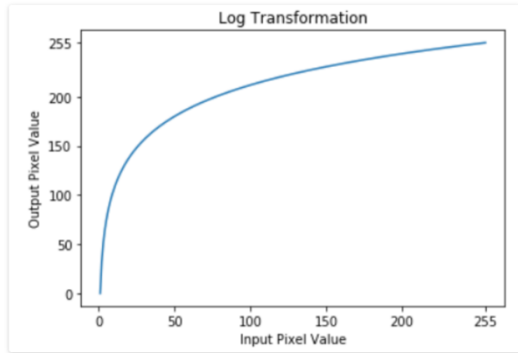


# Log Transformations

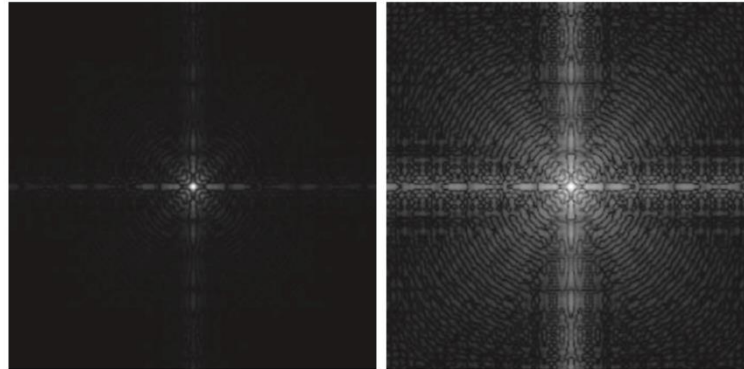
The general form of the log transformation is

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where  $c$  is a constant and it is assumed that  $r \geq 0$



- Expands the dark pixels in the image while compressing the brighter pixels
- Compresses the dynamic range (display of Fourier transform).

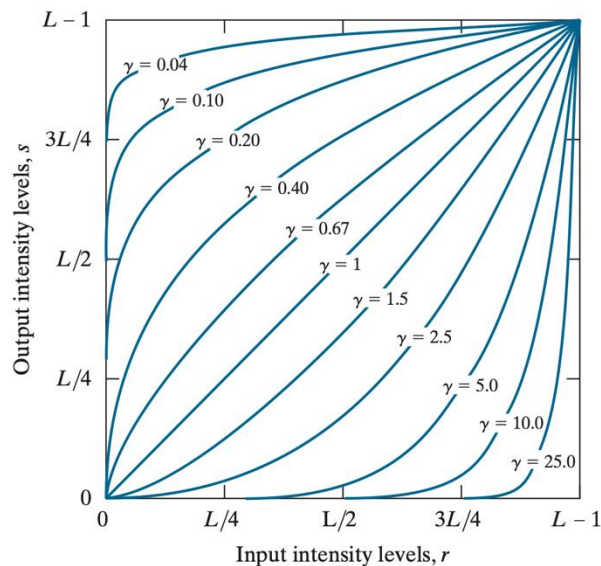


# Power-Law (Gamma) Transformations

Power-law transformations have the form

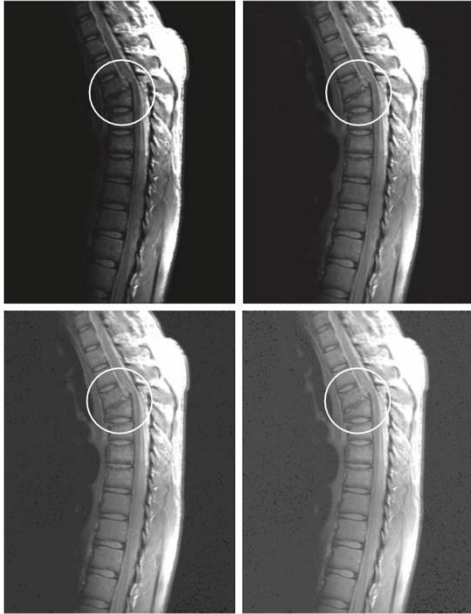
$$s = cr^\gamma$$

where  $c$  and  $\gamma$  are positive constants.





# Power-Law (Gamma) Transformations



(a) Magnetic resonance image (MRI) of a fractured human spine (the region of the fracture is enclosed by the circle).

(b)–(d) Results of applying the transformation in Eq. (3-5) with  $c = 1$  and  $g = 0.6, 0.4$ , and  $0.3$ , respectively.

# Power-Law (Gamma) Transformations

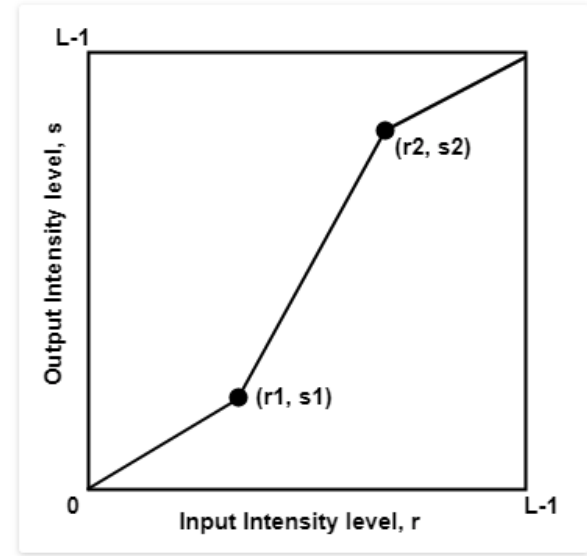


- a) Aerial image.
- b) (b)–(d) Results of applying the transformation in Eq. (3-5) with  $g = 3.0, 4.0, \text{ and } 5.0$ , respectively. ( $c = 1$  in all cases.) (Original image courtesy of NASA.)

# Contrast Stretching

By changing the location of points  $(r_1, s_1)$  and  $(r_2, s_2)$ , we can control the shape of the transformation function.

1. When  $r_1 = s_1$  and  $r_2 = s_2$ , transformation becomes a **Linear function**.
2. When  $r_1 = r_2$ ,  $s_1 = 0$  and  $s_2 = L-1$ , transformation becomes a **Thresholding function**.
3. When  $(r_1, s_1) = (r_{\min}, 0)$  and  $(r_2, s_2) = (r_{\max}, L-1)$ , this is known as **Min-Max Stretching**.
4. When  $(r_1, s_1) = (r_{\min} + c, 0)$  and  $(r_2, s_2) = (r_{\max} - c, L-1)$ , this is known as **Percentile Stretching**.

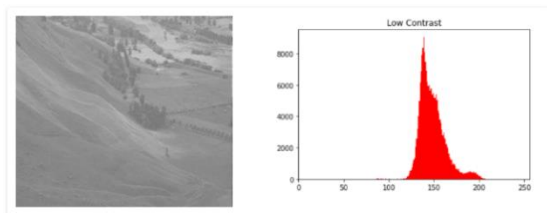


# Contrast Stretching

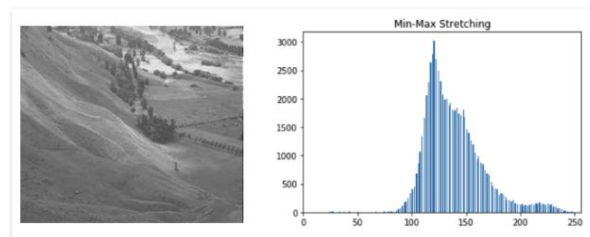
In **Min-Max Stretching**, the lower and upper values of the input image are made to span the full dynamic range. In other words, Lower value of the input image is mapped to 0 and the upper value is mapped to 255. All other intermediate values are reassigned new intensity values according to the following formulae

$$X_{new} = \frac{X_{input} - X_{min}}{X_{max} - X_{min}} \times 255$$

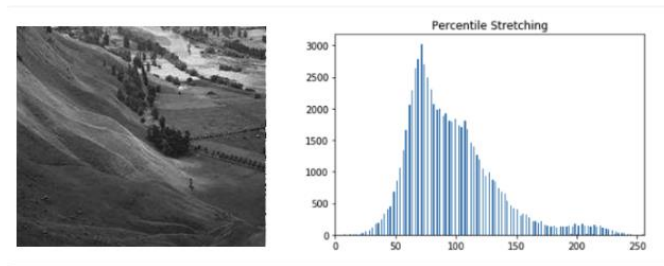
# Contrast Stretching



Original Image



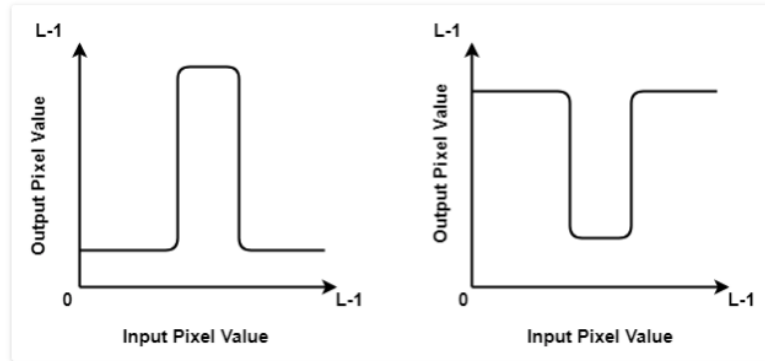
Min-max Stretching



Percentile Stretching

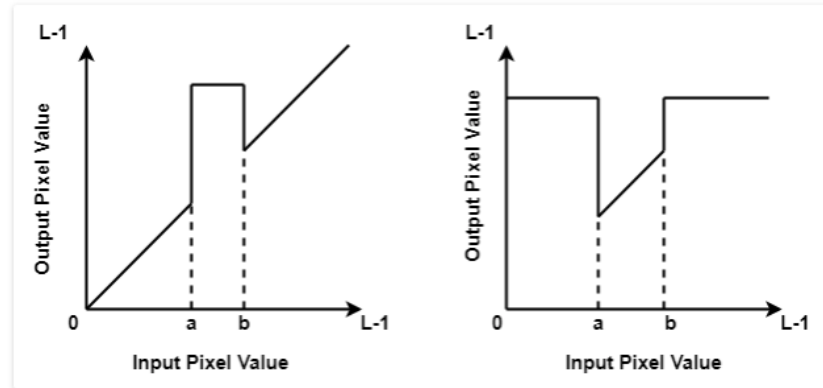
# Intensity-level Slicing

In the first type, we display the desired range of intensities in white and suppress all other intensities to black or vice versa. This results in a binary image. The transformation function for both the cases is shown below.

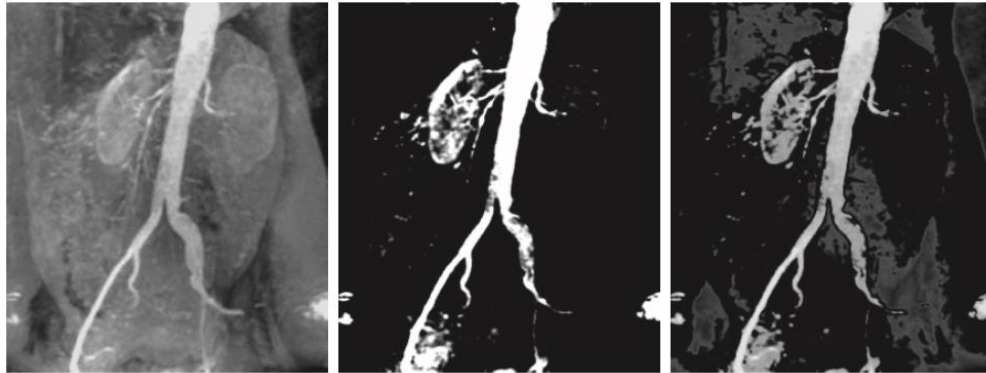


# Intensity-level Slicing

In the second type, we brighten or darken the desired range of intensities (a to b as shown below) and leave other intensities unchanged or vice versa. The transformation function for both the cases, first where the desired range is changed and second where it is unchanged, is shown below.



# Intensity-level Slicing



(a) Aortic angiogram.

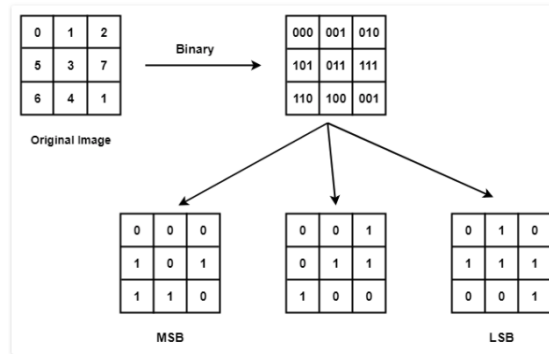
(b) Result of using a slicing transformation with the range of intensities of interest selected in the upper end of the gray scale.

(c) Result of using the transformation with the selected range set near black, so that the grays in the area of the blood vessels and kidneys were preserved.

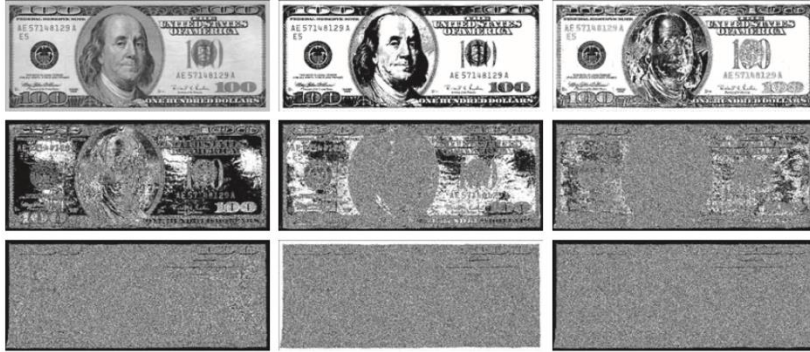


# Bit-plane Slicing

For an 8-bit image, a pixel value of 0 is represented as 00000000 in binary form and 255 is encoded as 11111111. Here, the leftmost bit is known as the most significant bit (MSB) as it contributes the maximum. e.g. if MSB of 11111111 is changed to 0 (i.e. 01111111), then the value changes from 255 to 127. Similarly, rightmost bit is known as Least significant bit (LSB).



# Bit-plane Slicing



(a) An 8-bit gray-scale image of size  $550 \times 1192$  pixels.

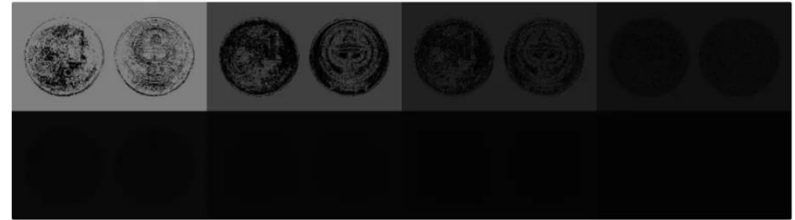
(b) through (i) Bit planes 8 through 1, with bit plane 1 corresponding to the least significant bit. Each bit plane is a binary image.



Image reconstructed from bit planes: (a) 8 and 7; (b) 8, 7, and 6; (c) 8, 7, 6, and 5.



Original Image



8 bit planes (Top row – 8,7,6,5 ; bottom – 4,3,2,1 bit planes)



Image using 4 bit planes (8,7,6,5)

# Bit-plane Slicing

Find all the bit planes of the following 4-bit image:

|   |    |    |    |
|---|----|----|----|
| 0 | 1  | 8  | 6  |
| 2 | 2  | 1  | 1  |
| 1 | 15 | 14 | 12 |
| 3 | 6  | 9  | 10 |

# Histogram Equalization

Let  $r_k$ , for  $k = 0, 1, 2, \dots, L - 1$ , denote the intensities of an  $L$ -level digital image,  $f(x, y)$ . The *unnormalized histogram* of  $f$  is defined as

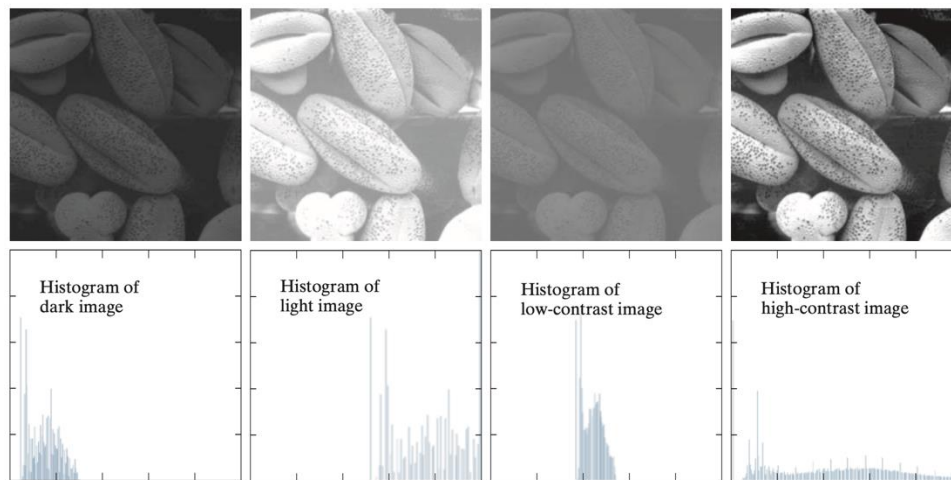
$$h(r_k) = n_k \quad \text{for } k = 0, 1, 2, \dots, L - 1$$

Where  $n_k$  is the number of pixels in  $f$  with intensity  $r_k$ , and the subdivisions of the intensity scale are called *histogram bins*. Similarly, the *normalized histogram* of  $f$  is defined as

$$p(r_k) = \frac{h(r_k)}{MN} = \frac{n_k}{MN}$$

where, as usual,  $M$  and  $N$  are the number of image rows and columns, respectively.

# Histogram Equalization



Four image types and their corresponding histograms. (a) dark; (b) light; (c) low contrast; (d) high contrast. The horizontal axis of the histograms are values of  $r_k$  and the vertical axis are values of  $p(r_k)$ .

# Histogram Equalization

Assuming initially continuous intensity values, let the variable  $r$  denote the intensities of an image to be processed. As usual, we assume that  $r$  is in the range  $[0, L - 1]$ , with  $r = 0$  representing black and  $r = L - 1$  representing white.

$$s = T(r) \quad 0 \leq r \leq L - 1$$

For  $r$  satisfying these conditions, we focus attention on transformations (intensity mappings) of the form that produce an output intensity value,  $s$ , for a given intensity value  $r$  in the input image. We assume that

- (a)**  $T(r)$  is a monotonic increasing function in the interval  $0 \leq r \leq L - 1$ ; and
- (b)**  $0 \leq T(r) \leq L - 1$  for  $0 \leq r \leq L - 1$ .

In some formulations to be discussed shortly, we use the inverse transformation

$$r = T^{-1}(s) \quad 0 \leq s \leq L - 1$$

# Histogram Equalization

The intensity of an image may be viewed as a random variable in the interval  $[0, L - 1]$ . Let  $p_r(r)$  and  $p_s(s)$  denote the PDFs of intensity values  $r$  and  $s$  in two different images. The subscripts on  $p$  indicate that  $p_r$  and  $p_s$  are different functions. A fundamental result from probability theory is that if  $p_r(r)$  and  $T(r)$  are known, and  $T(r)$  is continuous and differentiable over the range of values of interest, then the PDF of the transformed (mapped) variable  $s$  can be obtained as

$$p_s(s) = p_r(r) \left| \frac{dr}{ds} \right|$$

We define  $s$  or  $T(r)$  is

$$s = T(r) = (L - 1) \int_0^r p_r(w) dw$$

# Histogram Equalization

With Leibniz's rule we have:

$$\begin{aligned}\frac{ds}{dr} &= \frac{dT(r)}{dr} \\ &= (L-1) \frac{d}{dr} \left[ \int_0^r p_r(w) dw \right] \\ &= (L-1)p_r(r)\end{aligned}$$

$$\begin{aligned}p_s(s) &= p_r(r) \left| \frac{dr}{ds} \right| \\ &= p_r(r) \left| \frac{1}{(L-1)p_r(r)} \right| \\ &= \frac{1}{L-1} \quad 0 \leq s \leq L-1\end{aligned}$$



# Histogram Equalization

Suppose that the (continuous) intensity values in an image have the PDF

$$p_r(r) = \begin{cases} \frac{2r}{(L-1)^2} & \text{for } 0 \leq r \leq L-1 \\ 0 & \text{otherwise} \end{cases}$$

# Histogram Equalization

Suppose that the (continuous) intensity values in an image have the PDF

$$p_r(r) = \begin{cases} \frac{2r}{(L-1)^2} & \text{for } 0 \leq r \leq L-1 \\ 0 & \text{otherwise} \end{cases}$$

$$s = T(r) = (L-1) \int_0^r p_r(w) dw = \frac{2}{L-1} \int_0^r w dw = \frac{r^2}{L-1}$$

$$\begin{aligned} p_s(s) &= p_r(r) \left| \frac{dr}{ds} \right| = \frac{2r}{(L-1)^2} \left| \left[ \frac{ds}{dr} \right]^{-1} \right| \\ &= \frac{2r}{(L-1)^2} \left| \left[ \frac{d}{dr} \frac{r^2}{L-1} \right]^{-1} \right| = \frac{2r}{(L-1)^2} \left| \frac{(L-1)}{2r} \right| = \frac{1}{L-1} \end{aligned}$$

# Histogram Equalization

Find the transformation function that (when applied to the histogram-equalized intensities,  $s$ ) will produce an image whose intensity PDF is  $p_z(z) = 3z^2/(L-1)^3$  for  $0 \leq z \leq L-1$  and  $p_z(z) = 0$  for other values of  $z$ .

# Histogram Equalization

Recall that the probability of occurrence of intensity level  $r_k$  in a digital image is approximated by

$$p_r(r_k) = \frac{n_k}{MN}$$

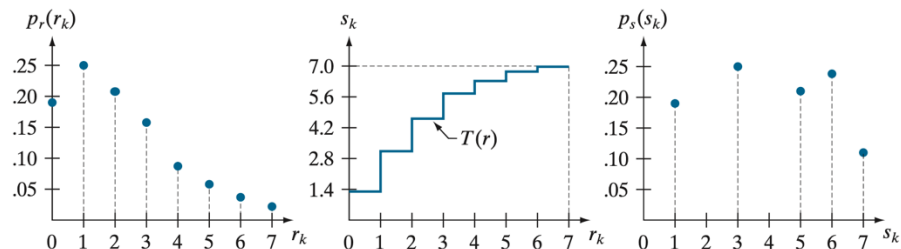
where  $MN$  is the total number of pixels in the image, and  $n_k$  denotes the number of pixels that have intensity  $r_k$ . As noted in the beginning of this section,  $p_r(r_k)$ , with  $r_k \in [0, L - 1]$ , is commonly referred to as a normalized image histogram.

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j) \quad k = 0, 1, 2, \dots, L - 1$$

where, as before,  $L$  is the number of possible intensity levels in the image

# Histogram Equalization

| $r_k$     | $n_k$ | $p_r(r_k) = n_k / MN$ |
|-----------|-------|-----------------------|
| $r_0 = 0$ | 790   | 0.19                  |
| $r_1 = 1$ | 1023  | 0.25                  |
| $r_2 = 2$ | 850   | 0.21                  |
| $r_3 = 3$ | 656   | 0.16                  |
| $r_4 = 4$ | 329   | 0.08                  |
| $r_5 = 5$ | 245   | 0.06                  |
| $r_6 = 6$ | 122   | 0.03                  |
| $r_7 = 7$ | 81    | 0.02                  |



It will be helpful to work through a simple example. Suppose that a 3-bit image ( $L = 8$ ) of size  $64 \times 64$  pixels ( $MN = 4096$ ) has the intensity distribution in the left table, where the intensity levels are integers in the range  $[0, L - 1] = [0, 7]$

$$\begin{array}{llll}
 s_0 = 1.33 \rightarrow 1 & s_2 = 4.55 \rightarrow 5 & s_4 = 6.23 \rightarrow 6 & s_6 = 6.86 \rightarrow 7 \\
 s_1 = 3.08 \rightarrow 3 & s_3 = 5.67 \rightarrow 6 & s_5 = 6.65 \rightarrow 7 & s_7 = 7.00 \rightarrow 7
 \end{array}$$

# Histogram Equalization

|   |   |   |   |
|---|---|---|---|
| 7 | 9 | 7 | 5 |
| 3 | 9 | 7 | 3 |
| 5 | 1 | 9 | 1 |
| 3 | 5 | 7 | 9 |

|    |    |    |    |
|----|----|----|----|
| 51 | 53 | 51 | 55 |
| 51 | 57 | 57 | 59 |
| 53 | 59 | 59 | 61 |
| 61 | 63 | 61 | 63 |

$L = [0, 255]$

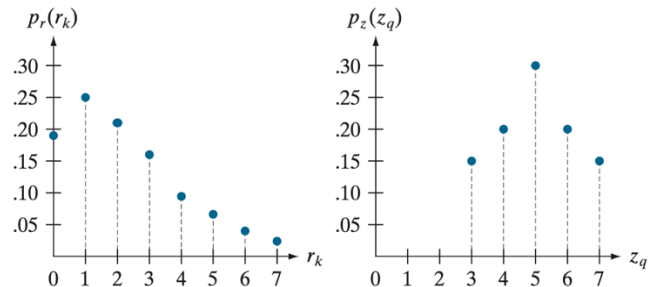
# Histogram Matching

$$s_k = T(r_k) = (L-1) \sum_{j=0}^k p_r(r_j) \quad k = 0, 1, 2, \dots, L-1$$

$$G(z_q) = (L-1) \sum_{i=0}^q p_z(z_i)$$

for a value of  $q$  so that

$$G(z_q) = s_k$$



Where  $p_z(z_i)$  is the  $i$ th value of the specified histogram. Finally, we obtain the desired value  $z_q$  from the inverse transformation:

$$z_q = G^{-1}(s_k)$$

# Histogram Matching

$$s_0 = 1; \quad s_1 = 3; \quad s_2 = 5; \quad s_3 = 6; \quad s_4 = 6; \quad s_5 = 7; \quad s_6 = 7; \quad s_7 = 7$$

$$G(z_0) = 0.00 \quad G(z_2) = 0.00 \quad G(z_4) = 2.45 \quad G(z_6) = 5.95$$

$$G(z_1) = 0.00 \quad G(z_3) = 1.05 \quad G(z_5) = 4.55 \quad G(z_7) = 7.00$$

$$G(z_0) = 0.00 \rightarrow 0$$

$$G(z_4) = 2.45 \rightarrow 2$$

$$G(z_1) = 0.00 \rightarrow 0$$

$$G(z_5) = 4.55 \rightarrow 5$$

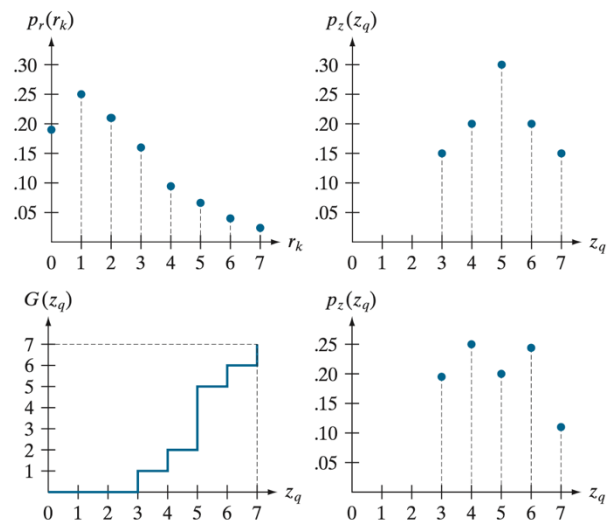
$$G(z_2) = 0.00 \rightarrow 0$$

$$G(z_6) = 5.95 \rightarrow 6$$

$$G(z_3) = 1.05 \rightarrow 1$$

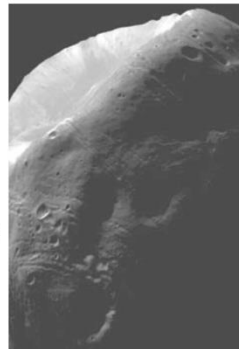
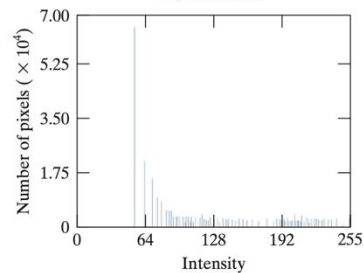
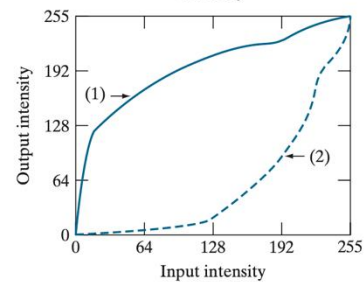
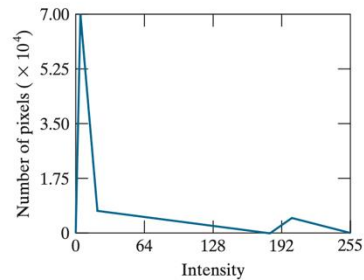
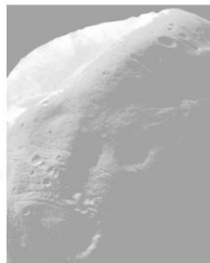
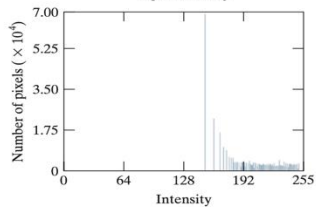
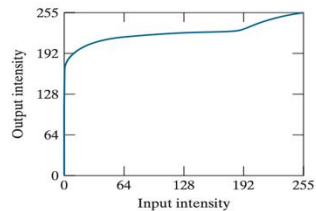
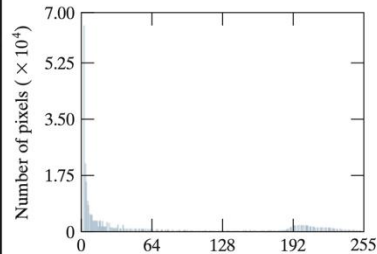
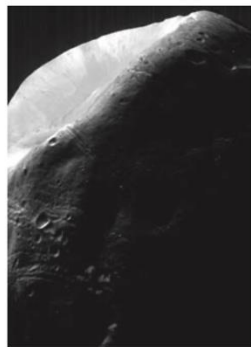
$$G(z_7) = 7.00 \rightarrow 7$$

| $s_k$ | $\rightarrow$ | $z_q$ |
|-------|---------------|-------|
| 1     | $\rightarrow$ | 3     |
| 3     | $\rightarrow$ | 4     |
| 5     | $\rightarrow$ | 5     |
| 6     | $\rightarrow$ | 6     |
| 7     | $\rightarrow$ | 7     |





# Histogram Matching



# Mean and Variance

For an image with intensity levels in the range  $[0, L - 1]$ , the  $n$ th moment of  $r$  about its mean,  $m$ , is defined as

$$\mu_n = \sum_{i=0}^{L-1} (r_i - m)^n p(r_i)$$

where  $m$  is given by

$$m = \sum_{i=0}^{L-1} r_i p(r_i)$$

The mean is a measure of average intensity and the variance (or standard deviation,  $\sigma$ ), given by

$$\sigma^2 = \mu_2 = \sum_{i=0}^{L-1} (r_i - m)^2 p(r_i)$$

# Using Histogram Statistics For Image Enhancement

